2022

(CBCS)

(5th Semester)

EDUCATION

SIXTH PAPER

(Statistics in Education)

Full Marks : 75

Time : 3 hours

The figures in the margin indicate full marks for the questions

(SECTION : A—OBJECTIVE)

(*Marks* : 10)

Tick (✓) the correct answer in the brackets provided : 1×10=10

1. Statistics that aim at learning characteristics of the population from a sample is

(a) descriptive ( )

(b) inferential ( )

(c) estimation ( )

(d) hypothesis ( )

2. Which of the following is not a graphic representation?

(a) Table ( )

(b) Polygon ( )

(c) Pie diagram ( )

(d) Bar chart ( )

3. The most popular measure of central tendency is

(a) median ( )

(b) range ( )

(c) mean ( )

(d) mode ( )

4. Median is a point on a scale such that half the scores falls above

(a) 80% ( )

(b) 70% ( )

(c) 60% ( )

(d) 50% ( )

5. Measures of variability are also known as

(a) measures of dispersion ( )

(b) measures of standard value ( )

(c) measures of equal value ( )

(d) measures of error value ( )

6. The square root of the average of the squared deviations of each score from the mean of a distribution of scores is known as

(a) range ( )

(b) standard deviation ( )

(c) average deviation ( )

(d) quartile deviation ( )

7. In a normal distribution, mean 1 standard deviation includes

(a) 64·26% of cases ( )

(b) 68·26% of cases ( )

(c) 72·36% of cases ( )

(d) 94·44% of cases ( )

8. Frequency distribution having the same kurtosis as the normal distribution positive is

(a) kurtosis ( )

(b) platykurtic ( )

(c) leptokurtic ( )

(d) mesokurtic ( )

9. Product moment method of correlation is developed by

(a) Charles Spearman ( )

(b) William Stern ( )

(c) Karl Pearson ( )

(d) Skinner ( )

10. Coefficient of correlation ranges from

(a) –1 through 0 to +1 ( )

(b) 1 through 0 to +3 ( )

(c) 3 through 0 to +3 ( )

(d) 3 through 0 to +1 ( )

(SECTION : B—SHORT ANSWER)

(*Marks* : 15)

Write briefly on the following : 3×5=15

UNIT—I

1. Meaning of descriptive statistics

OR

2. Meaning of graphical representation of data

UNIT—II

3. Concept of central tendency

OR

4. Uses of mode

UNIT—III

5. Meaning of standard deviation

OR

6. Concept of average deviation

UNIT—IV

7. Meaning of kurtosis

OR

8. Concept of NPC

UNIT—V

9. Interpretations of coefficient of correlation

OR

10. Types of correlation

(SECTION : C—DESCRIPTIVE)

(*Marks* : 50)

Answer the following : 10×5=50

UNIT—I

1. *(a)* Define Statistics. Mention its advantages. 2+2=4

(b) Tabulate the following scores into a frequency distribution with a size of class interval of 5 : 6

76	40	60	62	63	69	71	59
78	49	47	45	44	64	62	65
79	66	61	67	68	68	60	87
80	85	70	69	68	65	66	72
82	50	73	51	55	52	57	53

OR

2. *(a)* What is Statistics? Mention any two limitations of statistics. 2+2=4

(b) The scores of a student in a class is given below :

Subject	*Marks*
Languages	55%
Mathematics	45%
Science	40%
Social Science	60%

Draw a pie diagram to represent his/her performance. 6

UNIT—II

3. *(a)* Explain the concept of mean with its uses. 4

(b) Calculate the median from the distribution of scores : 6

Scores	*f*
70–74	2
65–69	2
60–64	2
55–59	4
50–54	5
45–49	9
40–44	6
35–39	3
30–34	4
25–29	2
20–24	1
N	40

OR

4. *(a)* What is mode? Mention the limitation of median. 4

(b) Calculate the mean from the data given in Question No. **3** *(b)*. 6

UNIT—III

5. *(a)* Give the meaning of median. 3

(b) Compute the quartile deviation from the following distribution of scores : 7

Scores	*f*
70–79	2
60–69	3
50–59	5
40–49	9
30–39	6
20–29	4
10–19	1
N	30

OR

6. *(a)* Explain the concept of measures of variability. 3

(b) Compute the standard deviation (SD) from the distribution of scores given in Question No. **5** *(b)*. 7

UNIT—IV

7. What is normal distribution? Discuss the characteristics of normal distribution with suitable diagram. 3+7=10

OR

8. *(a)* Explain the concepts of skewness and kurtosis with suitable diagram. 6

(b) Mention the applications of normal distribution curve in the field of education. 4

UNIT—V

9. *(a)* Explain the concept of correlation. 3

(b) Compute the coefficient of correlation between Mathematics and Science test scores by rank difference method and interpret your result : 5+2=7

Students	:	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*I*	*J*
Mathematics	:	65	48	50	75	58	60	69	73	55	82
Science	:	71	47	58	60	65	53	58	45	50	65

OR

10. *(a)* Mention the uses of correlation. 3

(b) Compute the coefficient of correlation from the scores given above in Question No. **9** *(b)* by using product moment method and interpret your result. 5+2=7

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